

Poisson

```
data epileptic_w;
input ID TRT Age C0 C1 C2 C3 C4;
datalines;
      1      0      31      11      5      3      3      3
      2      0      30      11      3      5      3      3
.....
     59      1      37      12      1      4      3      2
;
run;
```

Offset set
 $\log(w)$

GEE vs GLMM

- **GEE** is like studying all patients in a hospital and saying, "On average, this new epilepsy medication reduces seizure rates by 20%." It gives a **broad, public health-focused conclusion** without worrying about individual differences.
- **GLMM** is like a doctor adjusting a patient's treatment plan based on their unique seizure patterns. Some patients might have **higher baseline seizure rates** than others, and GLMM accounts for these individual differences.

Interpretation of Treatment Effects

- **Using GEE:**
 - The estimated treatment effect represents the **average reduction in seizure counts across all patients**.
 - Example interpretation: *"On average, patients taking the new medication experience 20% fewer seizures compared to the control group."*
 - This is useful if we care about **overall public health impact** rather than individual predictions.
- **Using GLMM:**
 - The treatment effect accounts for **individual differences in seizure frequency** (some patients may have naturally higher or lower seizure rates).
 - Example interpretation: *"For a given patient, the medication reduces their expected seizure count by 20% compared to their untreated seizure count."*
 - This is useful if we want **individualized predictions** (e.g., estimating how a specific patient might respond).

```
data epileptic;
set epileptic_w;
array AC(1:5) C0-C4;
array Aweek(1:5) (0 2 4 6 8);
do i=1 to 5;
  Count = AC[i];
  week = Aweek[i];
  L_per = log(2);
  if i eq 1 then L_per=log(8);
  output;
end;
```

```
drop C0-C4 Aweek1 - Aweek5 i;
run;
```

$$\log\left(\frac{Y_{ij}}{P_{ij}}\right) = \beta_0 + \beta_1 X_{ij}$$

Before fitting the GEE model, we'll do some exploratory analysis to understand how to include continuous covariates—particularly **time (in weeks)**—in the model. Since we are using a Poisson model with a log link, it is natural to examine the relationship between **time and the log-rate of the outcome**.

Specifically, we'll compute and plot estimated log-rates (e.g., log of the mean count) across time. This helps us assess whether the relationship between time and the outcome appears approximately linear on the log scale, or whether we may need a more flexible specification (e.g., quadratic terms or categorical time).

The goal is to let the data guide whether a simple linear effect of time is appropriate in the log-rate model.

```
data epileptic;
set epileptic;
  rate = Count/exp(L_per);
  l_count = log(Count+1);
  l_rate = log((Count+1)/exp(L_per));
run;
```

per = 8

```
proc print data=epileptic (obs=10);
run;
```

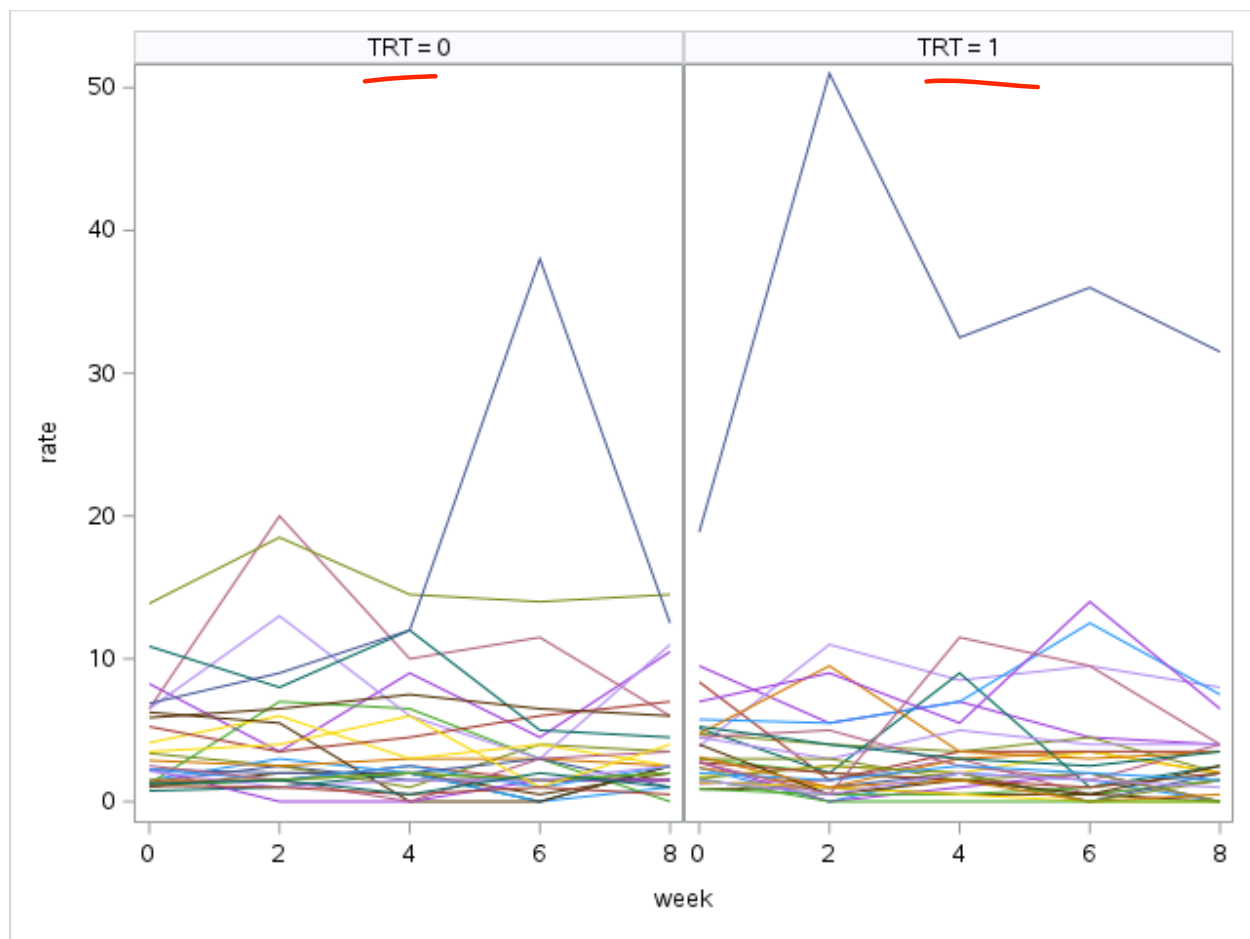
Obs	ID	TRT	Age	Count	week	L_per	rate	l_count	l_rate
1	1	0	31	11	0	2.07944	1.375	2.48491	0.40547
2	1	0	31	5	2	0.69315	2.500	1.79176	1.09861
3	1	0	31	3	4	0.69315	1.500	1.38629	0.69315
4	1	0	31	3	6	0.69315	1.500	1.38629	0.69315
5	1	0	31	3	8	0.69315	1.500	1.38629	0.69315
6	2	0	30	11	0	2.07944	1.375	2.48491	0.40547
7	2	0	30	3	2	0.69315	1.500	1.38629	0.69315
8	2	0	30	5	4	0.69315	2.500	1.79176	1.09861
9	2	0	30	3	6	0.69315	1.500	1.38629	0.69315

Obs	ID	TRT	Age	Count	week	L_per	rate	I_count	I_rate
10	2	0	30	3	8	0.69315	1.500	1.38629	0.69315

```

Proc SGpanel data = epileptic;
PanelBy TRT / columns=2;
series y=rate x=week / group =ID LineAttrs= (pattern=1 );
run;
quit;

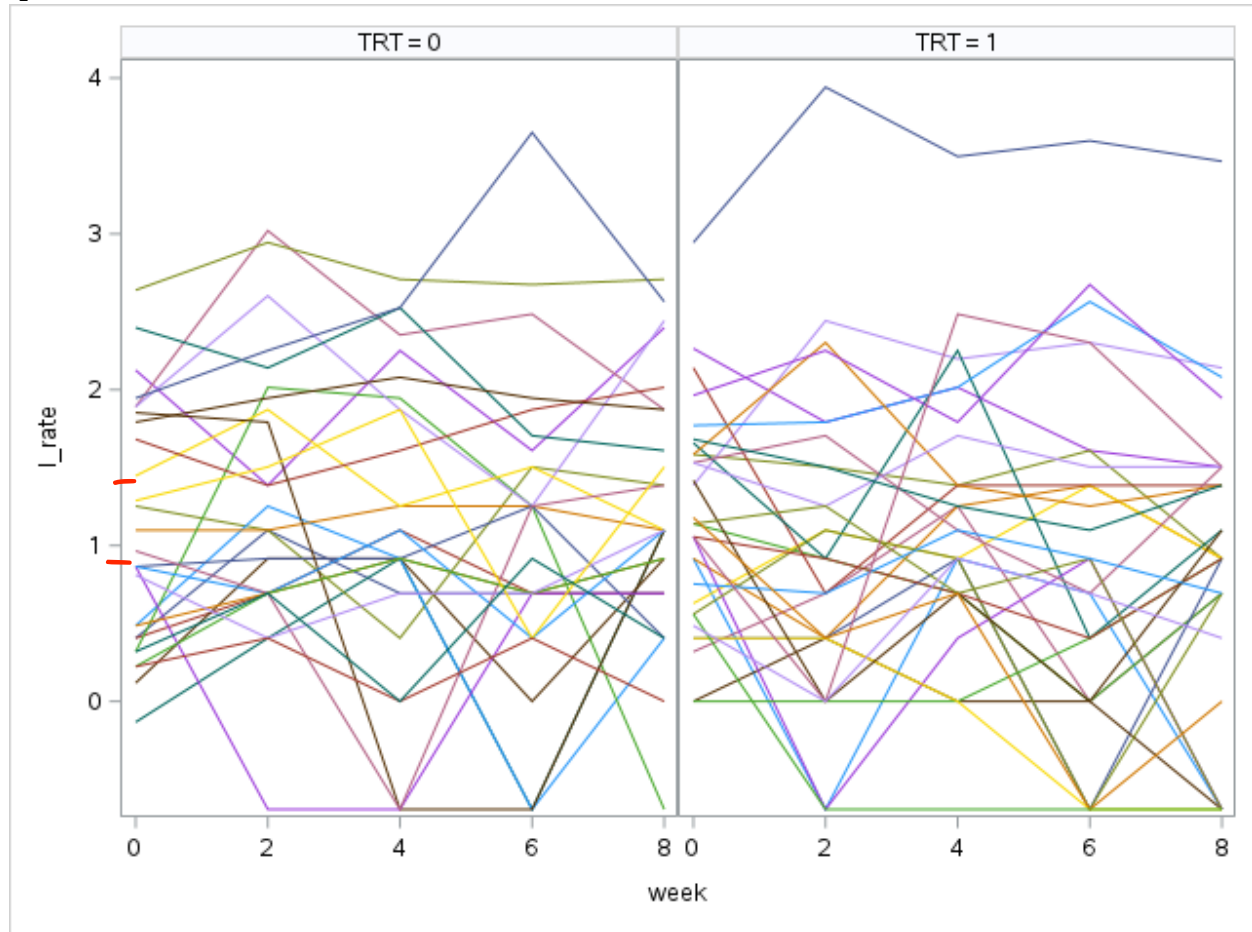
```



```

Proc SGpanel data = epileptic;
PanelBy TRT / columns=2;
series y=l_rate x=week / group =ID LineAttrs= (pattern=1 );
run;
quit;

```



```

proc sort data=epileptic;
by TRT week;

*Calculate the mean by week;
proc means mean data=epileptic noprint;
by TRT week;
var l_rate;
output out = MN_TRT_dat mean = mn_TRT_rate;
run;

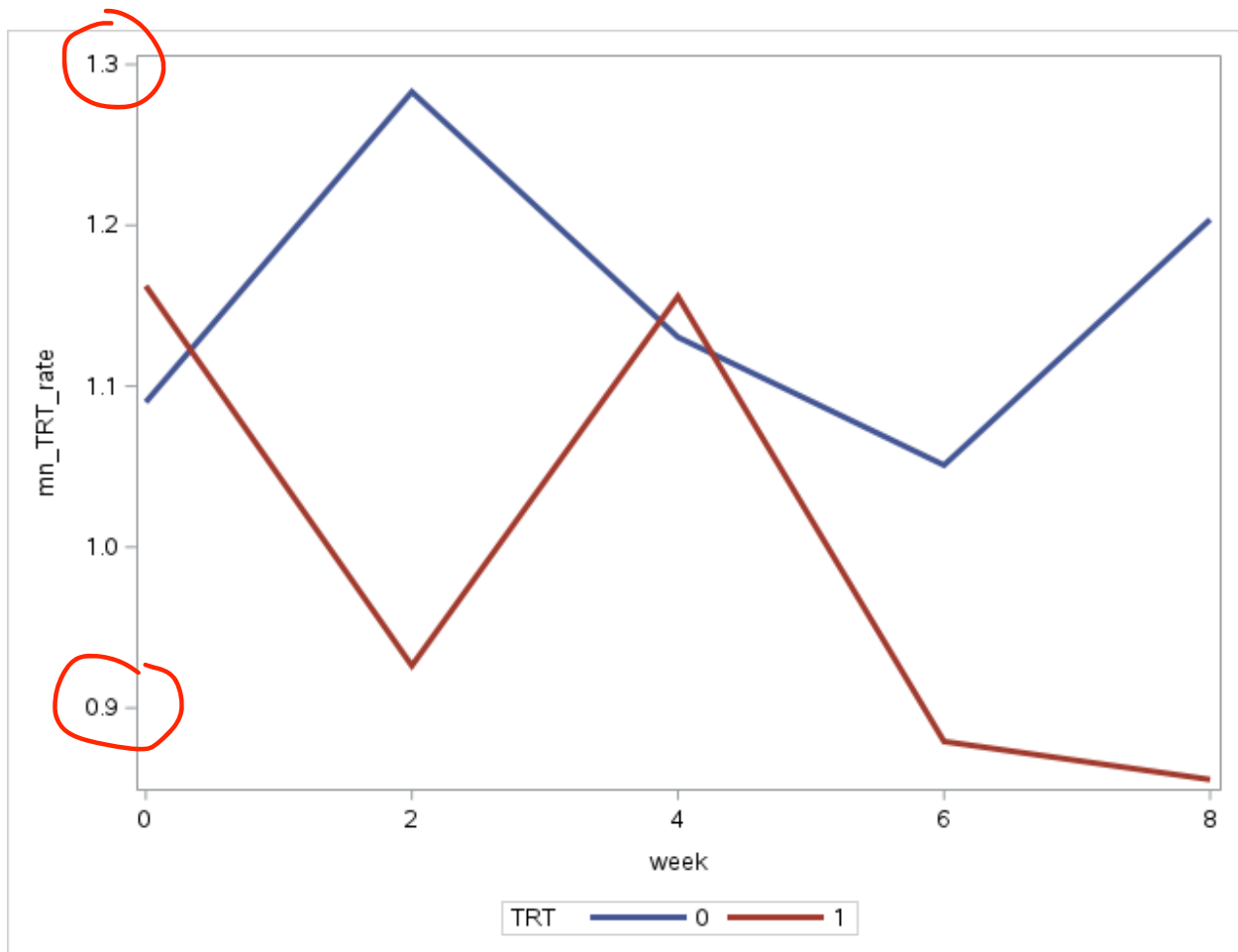
```

***First, let's look at the mean by TRT group;**

```

Proc SGplot data = MN_TRT_dat;
series x=week y=mn_TRT_rate / group =TRT LineAttrs= (pattern=1
thickness=3);
run;

```



```

proc gee data=epileptic;
class ID ;
model Count = TRT week TRT*week/d=poisson link=log offset=L_per
type3;
repeated subject=ID/type=exch corrw modelse covb ;
run;
quit;

```

The GEE Procedure

Model Information	
Data Set	WORK.EPILEPTIC
Distribution	Poisson
Link Function	Log
Dependent Variable	Count
Offset Variable	L_per
Number of Observations Read	295
Number of Observations Used	295

Class Level Information

Class	Levels	Values
ID	59	1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59

Parameter Information	
Parameter	Effect
Prm1	Intercept
Prm2	TRT
Prm3	week
Prm4	TRT*week

GEE Model Information	
Correlation Structure	Exchangeable
Subject Effect	ID (59 levels)
Number of Clusters	59
Correlation Matrix Dimension	5
Maximum Cluster Size	5
Minimum Cluster Size	5

Covariance Matrix (Empirical)				
	Prm1	Prm2	Prm3	Prm4
Prm1	0.02749	-0.02749	0.0001172	-0.000117
Prm2	-0.02749	0.04355	-0.000117	0.0003482
Prm3	0.0001172	-0.000117	0.0005812	-0.000581
Prm4	-0.000117	0.0003482	-0.000581	0.0008272

Working Correlation Matrix					
	Obs 1	Obs 2	Obs 3	Obs 4	Obs 5
Obs 1	1.0000	0.7781	0.7781	0.7781	0.7781
Obs 2	0.7781	1.0000	0.7781	0.7781	0.7781
Obs 3	0.7781	0.7781	1.0000	0.7781	0.7781
Obs 4	0.7781	0.7781	0.7781	1.0000	0.7781
Obs 5	0.7781	0.7781	0.7781	0.7781	1.0000

Exchangeable Working Correlation	
Correlation	0.7781

GEE Fit Criteria	
QIC	-581.1965
QICu	-581.7413



Parameter Estimates for Response Model						
with Empirical Standard Error Estimates						

Parameter	Estimate	Standard Error	95% Confidence Limits		Z	Pr > Z
Intercept	1.2901	0.1658	0.9651	1.6150	7.78	<.0001
TRT	0.0378	0.2087	-0.3712	0.4468	0.18	0.8563
week	-0.0009	0.0241	-0.0481	0.0464	-0.04	0.9708
TRT*week	-0.0188	0.0288	-0.0751	0.0376	-0.65	0.5143

Parameter Estimates for Response Model						
with Model-Based Standard Error Estimates						

Parameter	Estimate	Standard Error	95% Confidence Limits		Z	Pr > Z
Intercept	1.2901	0.1550	0.9862	1.5939	8.32	<.0001
TRT	0.0378	0.2123	-0.3783	0.4539	0.18	0.8587
week	-0.0009	0.0255	-0.0509	0.0491	-0.03	0.9724
TRT*week	-0.0188	0.0366	-0.0905	0.0529	-0.51	0.6081

Score Statistics For Type 3 GEE Analysis			
--	--	--	--

Source	DF	Chi-Square	Pr > ChiSq
TRT	1	0.03	0.8549
week	1	0.00	0.9680
TRT*week	1	0.46	0.4967

$$e^{-0.0188} =$$

Interpretation of interaction

Every additional week, the average seizure rate in the treatment group is multiplied by 0.9814 (or reduced by 1.86%) relative to the average seizure rate control group.

$$e^{0.04}$$

Interpretation of main effects

At the beginning of the study (week 0), the average seizure rate in the treatment group is 1.0385 times (or 3.85% higher than) the control group.

In the control group, each additional week the average seizure rate is multiplied by 0.9991 (or decreases by 0.09%).

$$e$$

QIC: What Is It?

QIC is an analogue of AIC for GEE models.

It is used to compare:

- Different **working correlation structures** (e.g., independence, exchangeable, AR(1))
- Sometimes different model specifications

Conceptually: $QIC = -2 Q(\text{independence model}) + 2 \text{ penalty}$

Where:

- Q is the quasi-likelihood evaluated under the **independence** working model
- The penalty reflects model complexity (similar to AIC)

Interpretation:

- **Smaller QIC = better model**
- Used primarily to choose the **working correlation structure**

QICu

QICu is a simplified version of QIC.

The “u” stands for “**uncorrected**” (or sometimes “under independence”).

$QIC = -2 Q(\text{independence model}) + 2 p$

Where:

- p = number of regression parameters

Key difference:

- QICu uses a simpler penalty (like AIC)
- QIC uses a more complex penalty involving the robust variance

When Do You Use Each?

Use QIC:

- When comparing different correlation structures
- This is the most common use in practice

Use QICu:

- When comparing different mean models (different covariates)
- Because the penalty is tied more directly to the number of parameters

```
proc gee data=epileptic;
class ID week (ref = '0');
model Count = TRT week TRT*week/d=poisson link=log offset=L_per
type3;
repeated subject=ID/type=exch corrw modelse covb ;
run;
quit;
```

GEE Fit Criteria	
QIC	-654.5784
QICu	-653.6516

Score Statistics For Type 3 GEE Analysis			
Source	DF	Chi-Square	Pr > ChiSq
TRT	1	0.03	0.8534
<u>week</u>	4	1.64	0.8012
TRT*week	4	1.80	0.7734

```
proc gee data=epileptic;
class ID ;
model Count = TRT week TRT*week/d=poisson link=log offset=L_per
type3;
repeated subject=ID/type=un corrw modelse covb ;
run;
quit;
```

Parameter Estimates for Response Model					
with Empirical Standard Error Estimates					
Parameter	Estimate	Standard Error	95% Confidence Limits		Z Pr > Z
Intercept	1.1074	0.3148	0.4904	1.7243	3.52 0.0004
TRT	0.1718	0.4013	-0.6148	0.9583	0.43 0.6687

Parameter Estimates for Response Model						
with Empirical Standard Error Estimates						
Parameter	Estimate	Standard Error	95% Confidence Limits		Z	Pr > Z
week	-0.0170	0.0658	-0.1460	0.1119	-0.26	0.7956
TRT*week	-0.0031	0.0742	-0.1486	0.1423	-0.04	0.9663

$$\text{Pois: } E(Y) = \text{Var}(Y)$$

```
proc gee data=epileptic;
class ID ;
model Count = TRT week TRT*week/d=negbin link=log offset=L_per
type3;
repeated subject=ID/type=exch corrw modelse covb ;
run;
quit;
```

The GEE Procedure		
Model Information		
Data Set	WORK.EPILEPTIC	
Distribution	Negative Binomial	
Link Function	Log	
Dependent Variable	Count	
Offset Variable	L_per	
Number of Observations Read	295	
Number of Observations Used	295	
Class Level Information		
Class	Levels	Values
ID	59	1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59

Parameter Information	
Parameter	Effect
Prm1	Intercept
Prm2	TRT
Prm3	week
Prm4	TRT*week

GEE Model Information	
Correlation Structure	Exchangeable
Subject Effect	ID (59 levels)
Number of Clusters	59
Correlation Matrix Dimension	5
Maximum Cluster Size	5
Minimum Cluster Size	5

Covariance Matrix (Model-Based)				
	Prm1	Prm2	Prm3	Prm4
Prm1	0.06288	-0.06288	-0.001524	0.001524
Prm2	-0.06288	0.11962	0.001524	-0.002855
Prm3	-0.001524	0.001524	0.0004494	-0.000449
Prm4	0.001524	-0.002855	-0.000449	0.0008587

Covariance Matrix (Empirical)				
	Prm1	Prm2	Prm3	Prm4
Prm1	0.02748	-0.02748	-0.000386	0.0003865

Covariance Matrix (Empirical)				
	Prm1	Prm2	Prm3	Prm4
Prm2	-0.02748	0.07472	0.0003865	0.001479
Prm3	-0.000386	0.0003865	0.0004222	-0.000422
Prm4	0.0003865	0.001479	-0.000422	0.0006613

Working Correlation Matrix					
	Obs 1	Obs 2	Obs 3	Obs 4	Obs 5
Obs 1	1.0000	0.7486	0.7486	0.7486	0.7486
Obs 2	0.7486	1.0000	0.7486	0.7486	0.7486
Obs 3	0.7486	0.7486	1.0000	0.7486	0.7486
Obs 4	0.7486	0.7486	0.7486	1.0000	0.7486
Obs 5	0.7486	0.7486	0.7486	0.7486	1.0000

Exchangeable Working Correlation	
Correlation	0.7486

GEE Fit Criteria	
QIC	-8251.4951
QICu	-8258.4515

Parameter Estimates for Response Model						
with Empirical Standard Error Estimates						
Parameter	Estimate	Standard Error	95% Confidence Limits		Z	Pr > Z
Intercept	1.4214	0.1658	1.0965	1.7463	8.57	<.0001
TRT	0.0183	0.2733	-0.5175	0.5540	0.07	0.9467

Parameter Estimates for Response Model						
with Empirical Standard Error Estimates						
Parameter	Estimate	Standard Error	95% Confidence Limits		Z	Pr > Z
week	0.0009	0.0205	-0.0393	0.0412	0.05	0.9633
TRT*week	-0.0187	0.0257	-0.0691	0.0317	-0.73	0.4660
Parameter Estimates for Response Model						
with Model-Based Standard Error Estimates						
Parameter	Estimate	Standard Error	95% Confidence Limits		Z	Pr > Z
Intercept	1.4214	0.2508	0.9300	1.9129	5.67	<.0001
TRT	0.0183	0.3459	-0.6596	0.6962	0.05	0.9579
week	0.0009	0.0212	-0.0406	0.0425	0.04	0.9645
TRT*week	-0.0187	0.0293	-0.0762	0.0387	-0.64	0.5224
Score Statistics For Type 3 GEE Analysis						
Source	DF	Chi-Square	Pr > ChiSq			
TRT	1	0.00	0.9477			
week	1	0.00	0.9644			
TRT*week	1	0.50	0.4779			

Count outcome:

- offset (different period lengths)
- mean of the log-rate by any continuous covariates.
- exponentiate coefficients (%)
- Poisson or negative binomial?